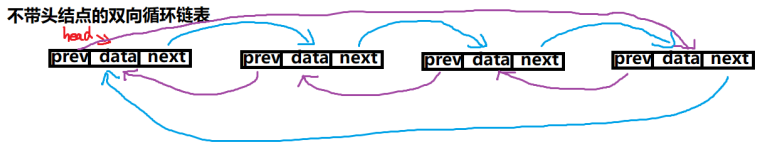

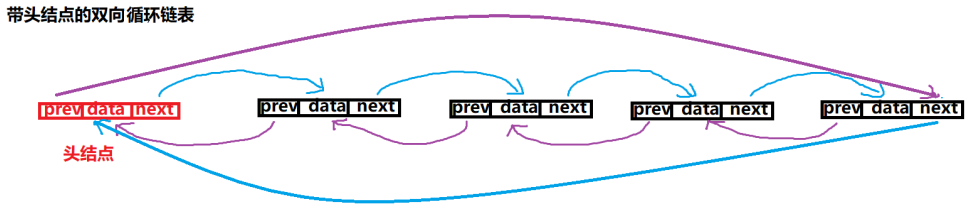


```
typedef int delemtype;
typedef struct dnode
{
    delemtype data;//数据域
    struct dnode *next;//指针域 保存下一个结点的地址
    struct dnode *prev;//指针域 保存上一个结点的地址
}dnode;
```

只有一个头结点

```
head->next = head;
head->prev = head;
```

双向循环链表怎么判断遍历结束条件?

```
dnode *p=head->next;
方法1:if(p == head) //循环结束
方法2：可以用结点的个数判断，head->data(可以存储结点的数目的)
```

双向循环链表怎么获取最后一个结点?

```
单向循环
dnode *p = head;
while(p->next!=head)
{
    p = p->next;
}
//跳出来的时候p就是最后一个结点

双向循环的最后一个结点 head->prev
```

创建一个头结点

```
//初始化双向循环链表  没有数据结点 只有一个头结点
dnode *init_circle_list()
{
    dnode *head = malloc(sizeof(dnode));
    head->data = 0;
    head->next = head->prev = head;
}
```

尾插

```
1.找最后一个结点 tail = head->prev

2.尾插  tail pnew head
    pnew->next = head;
    pnew->prev = tail;
    head->prev = pnew;
    tail->next = pnew;
```

头插

```
pnew->next = head->next;
pnew->prev = head;
head->next->prev = pnew;
head->next = pnew;
```

有序插入

```
//找
dnode *p = head->next;
dnode *pre = head;
while(p!=head)
{

}

//中间插  pre pnew  p
pnew->next = p;
pnew->prev = pre;
pre->next = pnew;
p->prev = pnew;
```

删除值为x的结点

```
//找
dnode *p = head->next;
dnode *pre = head;
while(p!=head)
{

}

//找到了
if(p!=head)
{
```

```
    //中间删除
    pre->next = p->next;
    p->next->prev = pre;
    //孤立p结点
    p->next = p->prev = NULL;
    free(p);
    p = NULL;
}
```

销毁

```
p = head->next;
whiel(p!=head)
{
    //循环头删
}

//删除头结点
```

circle_list.c/circle_list.h/main.c